

1 Three Dimensional Coordinate Systems and Vectors

Example 1.5.1 Find the equation of the plane which passes through the point $(2, 4, -1)$ with normal vector $\vec{n} = \langle 2, 3, 4 \rangle$.

Example 1.5.2 Find the equation of the plane which passes through the points $P(1, 3, 2)$, $Q(3, -1, 6)$ and $R(5, 2, 0)$.

Example 1.5.3 Find the point at which the line $x = 2 + 3t$, $y = -4t$, $z = 5 + t$ intercepts the plane $4x + 5y - 2z = 18$.

Example 1.5.4 Find the angle between the two planes $x + y + z = 1$ and $x - 2y + 3z = 1$.

1.5.1 Distances

Example 1.5.5 Find the distance from the point $(4, 0, 1)$ to the plane $x - y + z = 1$.

Example 1.5.6 Find the distance between the two planes $10x + 2y - 2z = 5$ and $5x + y - z = 1$.

1.6 Cylinders and Quadric Surfaces

Definition 1.6.1 A cylinder is a surface that consists of all lines that are parallel to a given line and pass through a given plane curve.

Definition 1.6.2 A quadric surface is the graph of a second-degree equation in three variables.

Example 1.6.3 Use traces to sketch the quadric surface with equation

$$x^2 + \frac{y^2}{9} + \frac{z^2}{4} = 1.$$

Example 1.6.4 Sketch the surface

$$z = 4x^2 + y^2.$$

Example 1.6.5 Sketch the surface

$$\frac{x^2}{4} + y^2 - \frac{z^2}{4} = 1.$$